

Computer Vision Practical Report:

PERSPECTIVE TRANSFORMATION

1. AIM

To read an input image using OpenCV, apply Perspective Transformation using four reference points to change the perspective view of the image, display the original and transformed images side-by-side using Matplotlib, and save the result to disk.

2. PROCEDURE

- Load the input image and select four source points representing the corners of the region to transform.
- Define four corresponding destination points to specify the desired rectangular output view.
- Calculate the 3×3 perspective transformation matrix using `cv2.getPerspectiveTransform()`.
- Apply the transformation to the image using `cv2.warpPerspective()`.
- Display the original and transformed images side-by-side using Matplotlib.
- Save the output image to disk using `cv2.imwrite()`.

3. PROGRAM CODE

```
import cv2
import numpy as np

# Read image
img = cv2.imread("E12-IMG.jpg")

if img is None:
    print("Error: Image not found!")
    exit()
```

```
# Resize image to small size
img = cv2.resize(img, (500, 400))

# Get image dimensions
h, w = img.shape[:2]

# Select four points from the original image
pts1 = np.float32([
    [50, 50],
    [450, 50],
    [50, 350],
    [450, 350]
])

# Corresponding points for perspective transformation
pts2 = np.float32([
    [80, 80],
    [420, 50],
    [50, 350],
    [450, 300]
])

# Calculate transformation matrix
M = cv2.getPerspectiveTransform(pts1, pts2)

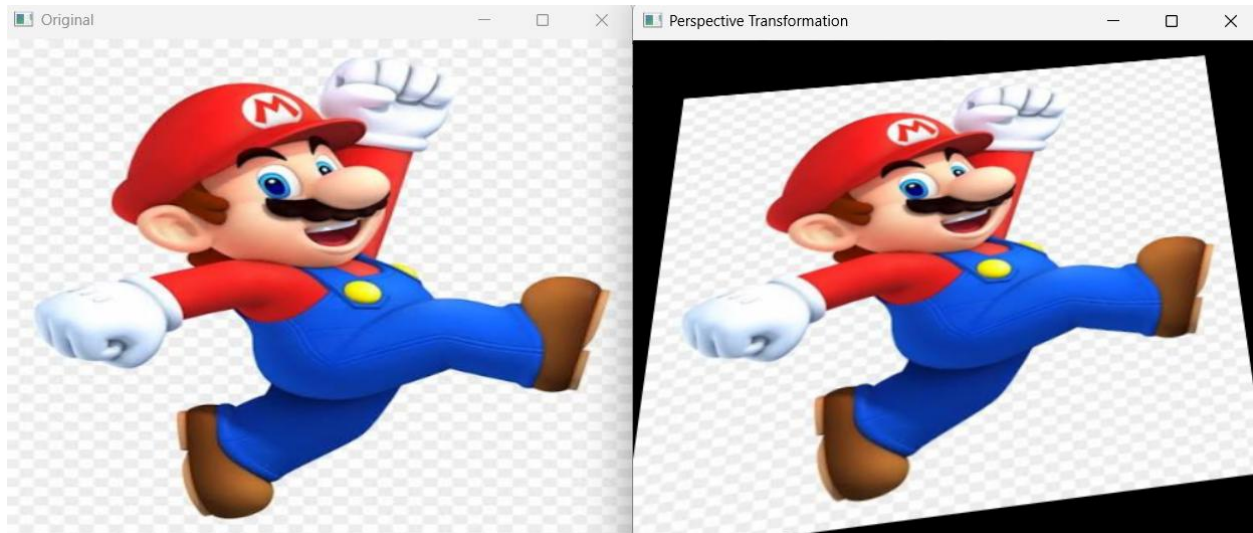
# Apply perspective transformation
output = cv2.warpPerspective(img, M, (w, h))

# Display images
cv2.imshow("Original", img)
cv2.imshow("Perspective Transformation", output)

# Save output
cv2.imwrite("perspective_output.jpg", output)
```

```
cv2.waitKey(0)  
cv2.destroyAllWindows()
```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image was loaded.